

Design of a Modular Smart Pill Dispenser for Reliable Medication Adherence Monitoring

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Abstract—

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I. INTRODUCTION

Population ageing is increasing the number of people living with chronic conditions and requiring long-term healthcare. A 2025 review reveals that about half of older adults experience multimorbidity, i.e. suffer from multiple chronic conditions simultaneously [1]; managing these conditions usually involves complex and long-term medication regimens, whose effectiveness depends in part on patients taking their medications as prescribed.

Medication adherence can be formally defined as “the process by which patients take their medications as prescribed, composed of initiation, implementation and discontinuation” [2]. [3] reports that between 44% and 76% of patients are non-adherent, with consequences that can be especially severe for elderly patients, possibly resulting in negative health outcomes such as increased hospitalization and mortality, as well as in increased healthcare costs [4]. Common causes of medication non-adherence include complexity of the prescribed treatment, forgetting or misunderstanding dosing instructions and even intentional opposition to treatment.

Several approaches have investigated smart medication and pill dispensers as a strategy to improve medication adherence by tackling these very reasons. According to [5], the majority of existing or proposed systems consist of:

- A health record system for hospitals
- A patient information system (used by caregivers to monitor the patient)
- A smart drug container
- A mobile tracking system, in the form of an application designed for patients

Most research prototypes use Arduino or Raspberry Pi as enabling technologies, with GSM, Wi-Fi or Bluetooth as pre-

ferred options for communication. Many smart pill dispensers also include sensors to detect the presence of pills and whether they have been removed or not, along with a physical interface to interact with the devices.

User studies of smart medication systems have reported high medication adherence during device usage [6], suggesting that this is the right direction to proceed; however, there are open issues and challenges that need to be addressed, mainly concerning hardware-software compatibility, a lack of authentication systems to prevent access to unauthorized users (like children) and general accessibility [5]. In particular, very few solutions adopt voice-controlled or easily customizable interfaces. Furthermore, modularity and context-adaptability seem to remain an open issue, with most of the available literature still at proposal stage.

The main contribution of this work is therefore to design and produce a prototype that detects whether and when the prescribed dose of a certain medication is removed from the corresponding compartment, while guiding the patient with progressive prompts and keeping the caregiver informed. In particular, the proposed design builds on the existing smart pill dispensers by explicitly considering modularity and simplicity of use as functional requirements.

The rest of the paper is organized as follows: section 2 presents a literature review, section 3 describes in detail our implementation of a smart pill dispenser according to the aforementioned criteria, section 4 provides the experimental results and section 5 discusses such results. Section 6 concludes the paper.

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